

# PRINCIPE D'HEISENBERG ET FONCTIONS POSITIVES THE HEISENBERG PRINCIPLE AND POSITIVE FUNCTIONS A BOURGAIN-PEDIA DIGESTION

J. BOURGAIN, L. CLOZEL AND J.-P. KAHANE

## §0. EDITORIAL PREFACE

This is the Bourgain-pedia *digestion* of Bourgain, Clozel and Kahane's *Principe d'Heisenberg et fonctions positives*: the paper at full length, with every step they compressed written out and every argument they omitted supplied.

**Source.** Unusually for this project, the source is the authors' own LaTeX, retrieved from the arXiv e-print service and kept byte-exact as `...-original.tex`. Nothing here is a transcription, so no formula below can have been corrupted by us in the reading; the slips listed are demonstrably theirs. The original is **19 pages**.

**Language.** The paper is in French; this digestion is in English, because the project is. Quotations are given in French with a translation. This is an editorial choice, not a rule of the blueprint — see `notes.md`.

**How to read the boxes.** Four kinds of material are ours, and each is boxed: *expansion* (a step of theirs, written out), *filled* (an argument absent from the original), *our notation*, and *caution* (something imported that we have not verified). Unboxed prose and displays are theirs.

**Everything numerical here has been recomputed independently**, before this text was drafted:  $\lambda$ ; both constants of Theorem 1; every coefficient of both Taylor expansions;  $P_4(3/2) = 3/2$ ;  $Q_4(c+1) = -c^2 + 1$ ; the  $a = \sqrt{2}$  and  $a = 2$  reductions;  $p_5 = 0$  and  $p_6 = -4/45$ ; and Theorem 3's inequality chain in nine dimensions. All agree with the paper.

**Seven slips in the original**, none affecting a theorem:

- (i) §1: “on en déduit que  $\mathcal{B}_1^- \leq B_1$ ” where  $\mathcal{B}_1^- \leq B_1^-$  is needed. [B20]
- (ii) §1: “ $B_1^- \leq (a\sqrt{2})^2 = 2a$ ” — the square is  $2a^2$ . [B27]
- (iii) §2:  $e^{X(2h-3h^2+3h^3-4h^4)X}$  carries a doubled  $X$ ; and the series is written  $P_1h + P_2h^2 + P_3h^3 + P_4h^5$ , where  $P_4h^4$  is meant. [C11]
- (iv) §2: the extremum condition is printed  $a(1-a) = 2\log(1+a)$ , which has no root  $> 1$ ; it must be  $a(a-1) = 2\log(1+a)$ , whose root *is* the 2.08137... printed. [C18]
- (v) §3: the label (3.4) is used twice, for two different equations. [D5]
- (vi) §4: Odlyzko's bound is written  $|D|^{1/d} \geq 22.2(1+O(d))$ ;  $o(1)$  is meant. [E21]
- (vii) §4: (4.9) typesets  $C = |D|^{-1/d_0}$ ; and the sentence after it cites “(3.10) et (4.8)” where (4.9) is meant. [E25, E26]

Two places where a stated bound is weaker than the paper's own method delivers — both flagged by the authors themselves as approximate, and both recorded in §5 rather than silently improved:

- §2: “une combinaison linéaire convenable de  $H_0$  et  $H_4$  donne  $\pi A^2 \leq 3$ ”. Carried out (§2.1, which the paper does not do), the combination gives  $\pi A^2 = 3/2$  — exactly the bound (2.5) obtained later by a different route. [C2]

- §2: the correction argument is stated to give  $A \leq 0.63\dots$ , “nous n’avons fait qu’un calcul très approché”. Optimised numerically it gives  $A \leq 0.5973$ . [C21]

**What we do not verify.** All of §4 rests on Tate’s thesis and on four further citations (Armitage, Serre, Odlyzko, Roquette/Golod–Shafarevič). We state each import precisely and check the paper’s use of it, but we have not read those sources; the **caution** boxes in §4 say so at each point.

## §0'. NOTATION

**Theirs.**

- $\widehat{f}(y) = \int f(x)e^{-2i\pi xy} dx$ , with inverse  $f(x) = \int \widehat{f}(y)e^{2i\pi xy} dy$ ; in  $\mathbb{R}^d$ ,  $xy$  becomes  $x \cdot y$  and  $dx$  Lebesgue measure. This convention makes  $\gamma(x) = e^{-\pi x^2}$  (resp.  $e^{-\pi \|x\|^2}$ ) self-dual, and every constant below depends on it. [A6]
- A *Fourier pair*  $(f, \widehat{f})$ : both in  $L^1$ , so both continuous and vanishing at infinity.
- $a_f, a_{\widehat{f}}$ : any thresholds beyond which  $f$ , resp.  $\widehat{f}$ , is  $\geq 0$ .
- $A(f) = \inf\{x > 0 : f \geq 0 \text{ on } (x, \infty)\}$  — the least such threshold; in  $\mathbb{R}^d$ ,  $\inf\{r > 0 : f(x) \geq 0 \text{ for } \|x\| > r\}$ .
- $f^\#$ : the spherical average of  $f$  (§3).

OUR NOTATION — not in the original

**The four constants**, which the paper introduces over two pages and never tabulates. All are infima over Fourier pairs satisfying (1–3);  $\mathcal{S}$  is the Schwartz class. [A2]

$$\begin{aligned} B_1 &= \inf A(f)A(\widehat{f}) \quad \text{over } f \in L^1 \\ \mathcal{B}_1 &= \inf A(f)A(\widehat{f}) \quad \text{over } f \in \mathcal{S} \\ B_1^- &= \inf A(f)A(\widehat{f}) \quad \text{over } f \in L^1 \text{ with } f(0) < 0 \\ \mathcal{B}_1^- &= \inf A(f)A(\widehat{f}) \quad \text{over } f \in \mathcal{S} \text{ with } f(0) < 0 \end{aligned}$$

and  $B_d, \mathcal{B}_d$  likewise in  $\mathbb{R}^d$ . §1 proves  $B_1 \leq \mathcal{B}_1 \leq 2B_1$ , so the two that matter differ by at most a factor 2. Note that in the source `\bb` and `\Bb` are both defined as  $\mathcal{B}$ , so  $\mathcal{B}_d$  appears under two spellings that typeset identically. [F1]

OUR NOTATION — not in the original

We write  $f^+ = \max(f, 0)$ ,  $f^- = \max(-f, 0)$ , so  $f = f^+ - f^-$  and  $|f| = f^+ + f^-$ ; the paper uses this without introducing it. We reserve  $\gamma$  for the Gaussian in whatever dimension is current. [F5]

## §0''. BACKGROUND

**Background** (Inversion and the elementary bounds). If  $f, \widehat{f} \in L^1(\mathbb{R}^d)$  then both are continuous and tend to 0 at infinity, and  $\|f\|_\infty \leq \|\widehat{f}\|_1$ ,  $\|\widehat{f}\|_\infty \leq \|f\|_1$ . In particular  $\widehat{f}(0) = \int f$  and  $f(0) = \int \widehat{f}$ .

**Background** (Dilation). For  $\lambda > 0$ , if  $g(x) = f(x/\lambda)$  then  $\widehat{g}(y) = \lambda^d \widehat{f}(\lambda y)$ . Hence  $A(g) = \lambda A(f)$  and  $A(\widehat{g}) = \lambda^{-1} A(\widehat{f})$ , so the product  $A(f)A(\widehat{f})$  is dilation-invariant.

**Background** (Hermite functions [C1]). There is an orthogonal basis  $(H_n)_{n \geq 0}$  of  $L^2(\mathbb{R})$  with  $H_n(x) = e^{-\pi x^2} P_n(x)$ ,  $\deg P_n = n$ , and  $\widehat{H}_n = i^{-n} H_n$  for the convention above. Consequently a self-dual  $f = \widehat{f}$  has an expansion supported on  $n \equiv 0 \pmod{4}$ .

**Background** (A function and its transform cannot both have compact support). If  $f \in L^1(\mathbb{R}^d)$  and both  $f$  and  $\widehat{f}$  have compact support then  $f = 0$ . (The transform of a compactly supported  $L^1$  function extends to an entire function on  $\mathbb{C}^d$ ; an entire function vanishing on a non-empty open set vanishes identically.)

**Background** (Stirling, lower bound). For  $m > 0$ ,  $\Gamma(m+1) > \sqrt{2\pi m} (m/e)^m$ .

## §1. THE PROBLEM, AND A LOWER BOUND FOR $B_1$

### OUR NOTATION — not in the original

**The paper's own summary**, in place of an abstract, translated. [A1] “We describe a natural problem concerning Fourier transforms, which introduces constants  $B_d$  depending on the dimension  $d$ . The problem is posed, in dimension 1, in the first section, where we describe a simple reduction to the case of self-dual functions. The first result is a lower bound for  $B_1$ . Then we consider a very natural class of functions which gives a simple upper bound for  $B_1$ . This is not optimal: we describe simple arguments which allow it to be improved (§2). In §3 these calculations are extended to arbitrary dimensions, giving simple lower and upper bounds for the constants. Finally §4, arithmetic, relates this problem to a well-known question concerning zeta functions of number fields. The arithmetic arguments show that the linear growth of  $B_d$  as a function of the dimension is natural in view of known properties of the ramification of these fields.”

**1.1 The conditions.** Consider a Fourier pair  $(f, \widehat{f})$  on the real line, and the conditions

- 1)  $f$  and  $\widehat{f}$  are real and even, not identically zero;
- 2)  $f(0) \leq 0$  and  $\widehat{f}(0) \leq 0$ ;
- 3)  $f(x) \geq 0$  for  $x \geq a_f$ , and  $\widehat{f}(y) \geq 0$  for  $y \geq a_{\widehat{f}}$ .

**Problem.** What is the infimum of the product  $a_f a_{\widehat{f}}$  over Fourier pairs satisfying (1–3)? Call it  $B_1 \geq 0$ .

### FILLED — an argument omitted in the original, supplied here

**Why  $a_f, a_{\widehat{f}} > 0$ .** [A3] The paper asserts that 2) and  $f \not\equiv 0$  force  $a_f, a_{\widehat{f}} > 0$ , without argument. Suppose  $a_f = 0$  were admissible, i.e.  $f \geq 0$  on all of  $(0, \infty)$ ;  $f$  is even, so  $f \geq 0$  everywhere except possibly at 0, and by continuity  $f \geq 0$  everywhere. Then  $\widehat{f}(0) = \int f \geq 0$ , and 2) gives  $\widehat{f}(0) \leq 0$ , so  $\int f = 0$  with  $f \geq 0$  continuous, whence  $f \equiv 0$  — excluded by 1). The same argument applies to  $\widehat{f}$ .

### FILLED — an argument omitted in the original, supplied here

**Such pairs exist, and the problem is not vacuous.** [A4] The paper says only “de tels couples existent à l’évidence”. Here is one. Take  $f(x) = x^2(x^2 - \frac{3}{2\pi})e^{-\pi x^2}$ . §2.1 below verifies that  $f = \widehat{f}$ , that  $f(0) = 0$ , and that  $f \geq 0$  exactly where  $x^2 \geq 3/(2\pi)$ . So  $(f, f)$  satisfies (1–3) with  $a_f = a_{\widehat{f}} = \sqrt{3/(2\pi)}$ , giving at once  $B_1 \leq 3/(2\pi) = 0.477\dots$

**And why positivity is not obvious.** [A5] The pair may be dilated:  $f(x/\lambda)$  has transform  $\lambda \widehat{f}(\lambda y)$ , which sends  $(a_f, a_{\widehat{f}}) \mapsto (\lambda a_f, \lambda^{-1} a_{\widehat{f}})$ . So each threshold separately can be made as small as one likes, and only the *product* is constrained. Nothing elementary prevents the product from being driven to 0 as well; that it cannot be is Theorem 1.

**1.2 Reduction to self-dual functions with  $f(0) = 0$ .** The product  $A(f)A(\widehat{f})$  is invariant under  $f(x) \mapsto f(x/\lambda)$ ,  $\widehat{f}(y) \mapsto \lambda \widehat{f}(\lambda y)$  (Background). Since  $B_1 = \inf A(f)A(\widehat{f})$ , one may restrict to pairs with  $A(f) = A(\widehat{f})$ . Then  $f + \widehat{f} \neq 0$ , and

$$A(f + \widehat{f}) \leq A(f) = A(\widehat{f}),$$

so  $B_1 = \inf A^2(f + \widehat{f})$ , and therefore

$$B_1 = A^2, \quad A = \inf A(f), \quad (1)$$

the infimum over  $f \in L^1(\mathbb{R})$  real, even, not identically zero, equal to its own Fourier transform, with  $f(0) \leq 0$ .

EXPANSION — a step they compressed, written out

**The three steps, done.** [B1, B2, B4, B5]

(a) *Normalising*  $A(f) = A(\widehat{f})$ . Given a pair, put  $\lambda = (A(\widehat{f})/A(f))^{1/2}$  and replace  $f(x)$  by  $f(x/\lambda)$ . By Background the new thresholds are  $\lambda A(f)$  and  $\lambda^{-1}A(\widehat{f})$ , and both equal  $\sqrt{A(f)A(\widehat{f})}$ . The product is unchanged, so the infimum is unchanged.

(b)  $f + \widehat{f}$  is admissible, and  $A(f + \widehat{f}) \leq A(f)$ . It is real and even;  $(f + \widehat{f})^\wedge = \widehat{f} + f$  because  $\widehat{\widehat{f}} = f$  for even  $f$ ; and  $(f + \widehat{f})(0) = f(0) + \widehat{f}(0) \leq 0$ . For  $x > A(f) = A(\widehat{f})$  both summands are  $\geq 0$ , so the sum is, giving  $A(f + \widehat{f}) \leq A(f)$ .

(c)  $f + \widehat{f} \not\equiv 0$ . The paper's parenthesis is "consider its values at points near  $A(f)$  and above it". [B3] In full: by definition of  $A(f)$  as an infimum,  $f$  takes a strictly negative value at some point arbitrarily close to  $A(f)$  from below — otherwise  $f \geq 0$  on a larger interval and  $A(f)$  was not the infimum. Take  $x_0 < A(f)$  with  $f(x_0) < 0$ . If  $f + \widehat{f}$  vanished identically then  $\widehat{f} = -f$ , so  $\widehat{f}(x_0) = -f(x_0) > 0$ ; but also  $\widehat{f} = -f$  gives  $f = \widehat{\widehat{f}} = -\widehat{f} = f$ , no contradiction yet. Instead use positivity: for  $x > A(f)$ ,  $f(x) \geq 0$  and  $\widehat{f}(x) \geq 0$ , and  $f + \widehat{f} \equiv 0$  forces both to vanish on  $(A(f), \infty)$ . Then  $f$  vanishes on a half-line, and being even, on  $|x| > A(f)$ :  $f$  has compact support. The same holds for  $\widehat{f}$ , so  $f \equiv 0$  by Background — contradicting 1).

(d) *Both inequalities for  $B_1 = \inf A^2(f + \widehat{f})$ .* [B5] " $\leq$ ": each  $f + \widehat{f}$  arising from an admissible pair is itself admissible and self-dual, so  $A^2(f + \widehat{f}) = A(f + \widehat{f})A(\widehat{f + \widehat{f}})$  is one of the products over which  $B_1$  is an infimum. " $\geq$ ": by (b),  $A^2(f + \widehat{f}) \leq A(f)A(\widehat{f})$  for every normalised pair, so the infimum over the self-dual ones is at most  $B_1$ .

Setting  $\gamma(x) = e^{-\pi x^2}$ , so that  $\gamma = \widehat{\gamma}$ : if  $f(0) < 0$  then  $f - f(0)\gamma$  is not identically zero, satisfies the same conditions, and  $A(f - f(0)\gamma) \leq A(f)$ . Finally

$$A = \inf A(f), \quad (2)$$

the infimum being over  $f \in L^1(\mathbb{R})$  real, even, not identically zero, with  $f = \widehat{f}$  and  $f(0) = 0$ .

EXPANSION — a step they compressed, written out

**The Gaussian correction.** [B6] Since  $f(0) < 0$ ,  $-f(0) > 0$ , so  $f - f(0)\gamma = f + |f(0)|\gamma$  adds a positive multiple of a positive function. Hence: it is self-dual (both summands are); its value at 0 is  $f(0) - f(0)\gamma(0) = f(0) - f(0) = 0$ ; it is  $\geq f$  pointwise, so it is  $\geq 0$  wherever  $f$  is, giving  $A(f - f(0)\gamma) \leq A(f)$ ; and it is not identically zero, since otherwise  $f = f(0)\gamma$  would be everywhere  $\leq 0$ , contradicting 3) as in the argument for  $a_f > 0$ .

### 1.3 Theorem 1.

**Theorem 1.** Let  $\lambda = -\inf\left(\frac{\sin x}{x}\right) = 0,2172\dots$ . Then

$$A \geq \frac{1}{2(1+\lambda)} = 0,4107\dots, \quad \text{hence} \quad B_1 \geq 0,1687\dots$$

**EXPANSION** — a step they compressed, written out

**The constant.** [B8, B9]  $\lambda$  is attained at the first positive critical point of  $\sin x/x$ , i.e. the first positive root of  $\tan x = x$ . Recomputed:  $x = 4,4934094579$ ,  $\lambda = 0,2172336282$ , and then  $1/(2(1+\lambda)) = 0,4107674882$  and  $A^2 \geq 0,1687299294$ . Both agree with the printed values.

*Proof.* Choose  $f = \widehat{f}$  with  $f(0) = 0$  and  $\int_{\mathbb{R}} |f| = 1$ ; write  $A = A(f)$ . Put  $f = f^+ - f^-$ ,  $|f| = f^+ + f^-$ . Since  $\int_{\mathbb{R}} f = \widehat{f}(0) = 0$  we have  $\int f^+ = \int f^- = \int_{-A}^A f^- = \frac{1}{2}$ . Hence  $\int_{|x| \geq A} |f| = \int_{|x| \geq A} f^+ \leq \frac{1}{2}$ , so  $\int_{|x| \leq A} |f| \geq \frac{1}{2}$ . But  $|f(x)| \leq \int |\widehat{f}| = 1$ . Therefore  $2A \geq \frac{1}{2}$ , a first bound  $A \geq \frac{1}{4}$ . This argument extends to higher dimensions (§3).

In dimension 1 it can be refined as follows. Since  $f$  is self-dual,

$$\begin{aligned} f(x) &= \int f(y) \cos 2\pi y x \, dy = \int f(y) (\cos 2\pi y x - 1) \, dy \\ &= \int f^-(y) (1 - \cos 2\pi y x) \, dy - \int f^+(y) (1 - \cos 2\pi y x) \, dy. \end{aligned}$$

Both integrals being positive, this implies

$$f^-(x) \leq \int f^+(y) (1 - \cos 2\pi y x) \, dy,$$

whence

$$\frac{1}{4} = \int_0^A f^- \leq \int_{-\infty}^{+\infty} f^+(y) \left[ A - \frac{\sin 2\pi y A}{2\pi y} \right] dy$$

and therefore

$$\frac{1}{4} \leq \frac{A}{2} \sup_{u \in \mathbb{R}} \left( 1 - \frac{\sin u}{u} \right) = \frac{A}{2} (1 + \lambda),$$

which is the theorem. □

**EXPANSION** — a step they compressed, written out

**Every step of that proof.** [B10–B18]

*Normalising*  $\int |f| = 1$  is legitimate:  $A(cf) = A(f)$  for  $c > 0$ , and  $f \mapsto cf$  preserves self-duality and  $f(0) = 0$ . [B10]

$\int f^+ = \int f^- = \frac{1}{2}$ . [B11] From  $\int f = \widehat{f}(0) = 0$  we get  $\int f^+ = \int f^-$ , and  $\int f^+ + \int f^- = \int |f| = 1$ . That  $\int f^-$  is concentrated on  $[-A, A]$  is because  $f \geq 0$  outside  $[-A, A]$  (evenness plus the definition of  $A$ ), so  $f^-$  vanishes there.

$\int_{|x| \leq A} |f| \geq \frac{1}{2}$ . [B12] On  $|x| \geq A$  we have  $f \geq 0$ , so  $|f| = f^+$  there, and  $\int_{|x| \geq A} f^+ \leq \int f^+ = \frac{1}{2}$ ; subtract from  $\int |f| = 1$ .

$\|f\|_{\infty} \leq 1$  and  $A \geq \frac{1}{4}$ . [B13] By Background,  $\|f\|_{\infty} \leq \|\widehat{f}\|_1 = \|f\|_1 = 1$  using  $f = \widehat{f}$ . Then  $\frac{1}{2} \leq \int_{|x| \leq A} |f| \leq 2A \cdot 1$ .

*The cosine identity.* [B15]  $f$  real and even gives  $\widehat{f}(x) = \int f(y) \cos 2\pi y x \, dy$ ; self-duality replaces  $\widehat{f}$  by  $f$  on the left. Subtracting  $\int f(y) \, dy = 0$  inserts the  $-1$ .

*The inequality for  $f^-$ .* [B16] Write  $I^\pm(x) = \int f^\pm(y)(1 - \cos 2\pi yx)dy$ , both  $\geq 0$  since  $1 - \cos \geq 0$  and  $f^\pm \geq 0$ . The identity is  $f = I^- - I^+$ , so  $-f = I^+ - I^- \leq I^+$ . Also  $0 \leq I^+$ . Hence  $f^- = \max(-f, 0) \leq I^+$ , which is the display.

*The Fubini step.* [B17]  $\int_0^A f^- = \frac{1}{4}$  because  $f^-$  is even with total mass  $\frac{1}{2}$ , supported in  $[-A, A]$ . Then

$$\int_0^A \int f^+(y)(1 - \cos 2\pi yx) dy dx = \int f^+(y) \left[ A - \frac{\sin 2\pi yA}{2\pi y} \right] dy,$$

the interchange being justified because the integrand is non-negative and  $f^+ \in L^1$  (Tonelli), and the inner integral is  $\int_0^A (1 - \cos 2\pi yx) dx = A - \sin(2\pi yA)/(2\pi y)$ .

*The last line.* [B18] Substituting  $u = 2\pi yA$ ,  $A - \sin(2\pi yA)/(2\pi y) = A(1 - \frac{\sin u}{u}) \leq A \sup_u (1 - \frac{\sin u}{u}) = A(1 + \lambda)$ , and  $\int f^+ = \frac{1}{2}$  gives the factor  $A(1 + \lambda)/2$ .

**1.4 The Schwartz class costs at most a factor 2.** Let  $\mathcal{B}_1 = A^2$  with  $A$  defined by (2) but with  $f \in \mathcal{S}$ . Evidently

$$B_1 \leq \mathcal{B}_1. \quad (3)$$

Let  $B_1^- = \inf(A^2 \mid f(0) < 0, f = \widehat{f} \text{ even} \neq 0, f \in L^1)$ , and  $\mathcal{B}_1^-$  likewise with  $f \in \mathcal{S}$ . Evidently

$$B_1^- \leq \mathcal{B}_1^-, \quad (4)$$

$$\mathcal{B}_1 \leq \mathcal{B}_1^-, \quad B_1 \leq B_1^-. \quad (5)$$

Let us check that  $\mathcal{B}_1^- \leq B_1^-$ . Let  $f \in L^1$  satisfy the condition of (2) but with  $f(0) < 0$ , and let  $a = A(f)$ . Let  $\varphi = \psi * \psi$  with  $\psi$  smooth, even, positive, of compact support very close to 0, and  $g = f * \varphi$ . Then  $A(g) \leq a + \varepsilon$  and  $g(0) < 0$ . We have  $\widehat{g} = \widehat{f} \widehat{\psi}^2$ ; performing the same operation on  $\widehat{g}$  we obtain  $h \in \mathcal{S}$  with  $h = \widehat{h}$ ,  $h(0) < 0$  and  $A(h) \leq a + \varepsilon$ ; whence  $\mathcal{B}_1^- \leq B_1^-$ , that is

$$\mathcal{B}_1^- = B_1^-. \quad (6)$$

**EXPANSION** — a step they compressed, written out

**Two things about that paragraph.** [B20, B21]

*The printed conclusion.* The original reads “on en déduit que  $\mathcal{B}_1^- \leq B_1$  soit  $\mathcal{B}_1^- = B_1^-$ ”. The intermediate inequality must be  $\mathcal{B}_1^- \leq B_1^-$ : it is  $B_1^-$ , not  $B_1$ , that combines with (4) to give the stated equality. We have written  $B_1^-$ .

*Why the operation is performed twice.* One convolution does not preserve self-duality:  $g = f * \varphi$  has  $\widehat{g} = \widehat{f} \widehat{\psi}^2$ , a *product*, not a convolution, so  $g \neq \widehat{g}$  in general. Repeating on the transform side — multiplying  $g$  by a suitable smooth positive even factor, which is convolution on the transform side — restores the symmetry. Concretely, with  $\psi$  real even one has  $\widehat{\psi}$  real even, so  $h := (f * \varphi) \cdot \widehat{\varphi}$  satisfies  $\widehat{h} = \widehat{f} \widehat{\varphi} \cdot \varphi = h$  using  $f = \widehat{f}$  and  $\varphi, \widehat{\varphi}$  real even. Both operations move  $A$  by at most  $\varepsilon$  if  $\text{supp } \psi$  is small enough, and neither changes the sign of the value at 0. That  $g(0) < 0$  survives is because  $\varphi \geq 0$  has mass 1 and  $f$  is continuous and negative near 0.

Note that the argument does not apply if  $f(0) = 0$ . We shall show

$$B_1^- \leq 2B_1; \quad (7)$$

by (5) and (7) we deduce

$$B_1 \leq \mathcal{B}_1 \leq 2B_1. \quad (8)$$

EXPANSION — a step they compressed, written out

**The chain, checked.** [B19, B28] The left inequality is (3). For the right:  $B_1 \leq B_1^-$  by (5),  $B_1^- = B_1^-$  by (6), and  $B_1^- \leq 2B_1$  by (7); compose. Note that (6) is the only one of the four relations requiring work — (3), (4) and (5) are immediate from the definitions, since each enlarges the class over which the infimum is taken or drops a constraint.

Let  $f$  satisfy (2) and  $a = A(f)$ . Since  $\hat{f}(0) = \int f = 0$ ,  $f$  takes strictly negative values on  $[-a, a]$ . Let  $b > 0$  with  $f(b) < 0$ , and consider the distribution

$$T = \delta_b + \delta_{-b} + 2\delta_0.$$

It is a positive measure, of positive type:

$$\hat{T} = 2 \cos(2\pi by) + 2 \geq 0.$$

We have  $(T * f)(0) = f(b) + f(-b) < 0$ . Since  $b < a$ ,  $g = T * f$  satisfies

$$g(0) < 0, \quad g \geq 0 \text{ on } [2a, \infty),$$

and moreover  $\hat{g} = \hat{T}\hat{f}$  is  $\geq 0$  on  $[a, \infty)$ , with  $\hat{g}(0) = 0$ . By dilation one then obtains  $h$  with

$$h \geq 0 \text{ on } [a\sqrt{2}, \infty), \quad h(0) < 0; \quad \hat{h} \geq 0 \text{ on } [a\sqrt{2}, \infty), \quad \hat{h}(0) = 0.$$

The functions  $h, \hat{h}$  are real and even. Then  $h + \hat{h}$  satisfies the conditions defining  $B_1^-$ . So  $B_1^- \leq (a\sqrt{2})^2 = 2a^2$ ; varying  $f$  gives (7).

EXPANSION — a step they compressed, written out

**The convolution trick, in full.** [B22–B27]

*T is of positive type.*  $\hat{\delta}_c(y) = e^{-2\pi icy}$ , so  $\hat{T} = e^{-2\pi iby} + e^{2\pi iby} + 2 = 2 \cos(2\pi by) + 2 \geq 0$ .

*The value at 0.*  $(T * f)(0) = f(-b) + f(b) + 2f(0)$ . The printed line drops the last term because  $f(0) = 0$  under (2); we say so rather than leaving the reader to notice. [B23]

*$g \geq 0$  on  $[2a, \infty)$ .* [B24]  $g(x) = f(x - b) + f(x + b) + 2f(x)$ . For  $x \geq 2a$  and  $0 < b < a$ , each of  $x - b, x, x + b$  is  $\geq 2a - a = a$ , so each term is  $\geq 0$ .

*The transform side.*  $\hat{g} = \hat{T}\hat{f}$  with  $\hat{T} \geq 0$  and  $\hat{f} \geq 0$  on  $[a, \infty)$ , so  $\hat{g} \geq 0$  there; and  $\hat{g}(0) = \hat{T}(0)\hat{f}(0) = 4 \cdot 0 = 0$ .

*The dilation.* [B26]  $g$  is non-negative from  $2a$  and  $\hat{g}$  from  $a$ : the two thresholds differ. Replace  $g(x)$  by  $g(x/\lambda)$ , whose transform is  $\lambda\hat{g}(\lambda y)$ ; the thresholds become  $2a\lambda$  and  $a/\lambda$ . They are equal when  $\lambda = 1/\sqrt{2}$ , and their common value is then  $2a/\sqrt{2} = a\sqrt{2}$ . This is the  $h$  of the display. Note  $h(0) = g(0) < 0$  and  $\hat{h}(0) = \lambda\hat{g}(0) = 0$ .

*Why  $h + \hat{h}$ .* It is self-dual, even, real;  $(h + \hat{h})(0) = h(0) < 0$ , so it is a competitor for  $B_1^-$ ; and both summands are  $\geq 0$  beyond  $a\sqrt{2}$ , so  $A(h + \hat{h}) \leq a\sqrt{2}$ .

*The printed value.* [B27]  $B_1^- \leq (a\sqrt{2})^2$  and  $(a\sqrt{2})^2 = 2a^2$ ; the original prints  $2a$ . It is  $2a^2$  that gives (7), since  $B_1 = \inf a^2$  over the competitors of (2).

## §2. AN UPPER BOUND FOR $B_1$

**2.1 The Hermite route, carried out.** A first idea is to associate to  $f$  its Hermite expansion  $f \sim \sum_0^\infty a_n H_n(x)$ , where the  $H_n$  are eigenvectors of the Fourier operator  $\mathcal{F}$  with eigenvalues  $i^n$ ; thus  $f = \hat{f}$  is expressed as  $f \sim \sum_0^\infty a_{4m} H_{4m}$ . The  $H_n$  have the form  $H_n(x) = e^{-\pi x^2} P_n(x)$  with  $P_n$  a polynomial of degree  $n$ . A suitable linear combination of  $H_0$  and  $H_4$  (such that  $f(0) = 0$ ) gives  $\pi A^2 \leq 3$ . Beyond this the calculations seem difficult and we have not pursued this route.



**FILLED** — an argument omitted in the original, supplied here

**The combination, and what it actually gives.** [C2] The paper states  $\pi A^2 \leq 3$  and does not compute. Carrying it out is short, and the answer is better than advertised.

Seek  $f = (\alpha + \beta x^2 + \delta x^4)\gamma$ ,  $\gamma(x) = e^{-\pi x^2}$  — the general even element of the span of  $H_0, H_2, H_4$ . From  $\widehat{x f} = \frac{i}{2\pi} \widehat{f'}$  one gets  $\widehat{x^2 \gamma} = -\frac{1}{4\pi^2} \gamma''$  and  $\widehat{x^4 \gamma} = \frac{1}{16\pi^4} \gamma''''$ , and differentiating  $\gamma' = -2\pi x \gamma$  repeatedly,

$$\gamma'' = (4\pi^2 x^2 - 2\pi)\gamma, \quad \gamma'''' = (16\pi^4 x^4 - 48\pi^3 x^2 + 12\pi^2)\gamma.$$

Hence

$$\widehat{x^2 \gamma} = \left(\frac{1}{2\pi} - x^2\right)\gamma, \quad \widehat{x^4 \gamma} = \left(x^4 - \frac{3}{\pi}x^2 + \frac{3}{4\pi^2}\right)\gamma,$$

so that

$$\widehat{f} = \left[\alpha + \frac{\beta}{2\pi} + \frac{3\delta}{4\pi^2} - \left(\beta + \frac{3\delta}{\pi}\right)x^2 + \delta x^4\right]\gamma.$$

Matching coefficients with  $f$ : the  $x^4$  terms agree automatically; the  $x^2$  terms give  $2\beta = -3\delta/\pi$ ; and the constant terms give  $\beta/(2\pi) + 3\delta/(4\pi^2) = 0$ , which is the *same* condition. So self-duality is exactly  $\beta = -\frac{3\delta}{2\pi}$ , with  $\alpha$  free, and  $f(0) = \alpha$ , so the normalisation  $f(0) = 0$  forces  $\alpha = 0$ . Taking  $\delta = 1$ ,

$$f(x) = x^2 \left(x^2 - \frac{3}{2\pi}\right) e^{-\pi x^2}$$

which is self-dual (verified numerically to 12 digits at five points), vanishes at 0, and is  $\geq 0$  exactly for  $x^2 \geq 3/(2\pi)$ . Hence

$$A \leq \sqrt{\frac{3}{2\pi}} = 0.6909883\dots, \quad \pi A^2 = \frac{3}{2}.$$

So the Hermite route gives  $\pi A^2 \leq \frac{3}{2}$ , not  $\leq 3$ : a factor 2 better than printed, and *exactly* the bound (2.5) that §2.2 will reach by a quite different construction. Either the printed 3 is a slip for 3/2, or a different normalisation of  $H_4$  is intended; with the paper's own Fourier convention the computation gives 3/2. We record both and change no theorem:  $\pi A^2 \leq 3$  is true. This is also the explicit pair promised in §1.1.

**2.2 The family  $g_a$ .** One may also consider the functions

$$g_a(x) = a\gamma(ax) + \gamma\left(\frac{x}{a}\right) - (1+a)\gamma(x), \quad a > 1, \quad (9)$$

which satisfy the conditions of (2). Then any expression of the form

$$\int_1^\infty g_a(x) d\tau(a) \quad (10)$$

where  $\tau$  is a measure on  $]1, \infty[$  such that the integral converges for every  $x$  and is  $\geq 0$  at infinity, is a candidate function. (It seems difficult to determine a simple characteristic property of  $\tau$  ensuring that (10) converges and is positive at infinity.)

**EXPANSION** — a step they compressed, written out

$g_a$  is admissible. [C4] With the paper's convention,  $\widehat{\gamma(a \cdot)}(y) = a^{-1}\gamma(y/a)$ . Hence

$$\widehat{g_a} = a \cdot a^{-1}\gamma(y/a) + a\gamma(ay) - (1+a)\gamma(y) = g_a(y),$$

so  $g_a$  is self-dual; and  $g_a(0) = a + 1 - (1+a) = 0$ . It is real, even and integrable, so it is a competitor in (2) provided it is not identically zero, which  $g_a''(0) \neq 0$  shows for  $a \neq 1$ .



We first study  $A(g_a)$ . It is convenient to set  $X = \pi x^2$  and  $G_a(X) = g_a(x)$ , so

$$G_a(X) = a e^{-a^2 X} + e^{-a^{-2} X} - (1 + a)e^{-X}.$$

Moreover

$$H_a(X) = e^X G_a(X) = a e^{(1-a^2)X} + e^{(1-a^{-2})X} - 1 - a \quad (11)$$

is a convex function satisfying

$$H_a(0) = 0, \quad H'_a(0) = -a^{-2}(a^2 - 1)(a^3 - 1) < 0$$

and tending to  $+\infty$  with  $X$ . It therefore admits a unique zero  $X_a > 0$ , and

$$A(g_a) = \sqrt{\frac{X_a}{\pi}}.$$

**EXPANSION** — a step they compressed, written out

**Convexity, the derivative, and uniqueness.** [C6–C10]  $H_a$  is a sum of two exponentials  $e^{cX}$  (with  $c = 1 - a^2 < 0$  and  $c = 1 - a^{-2} > 0$ ) and a constant, hence convex.  $H_a(0) = a + 1 - 1 - a = 0$ . Differentiating,  $H'_a(0) = a(1 - a^2) + (1 - a^{-2})$ ; and

$$-a^{-2}(a^2 - 1)(a^3 - 1) = -a^{-2}(a^5 - a^3 - a^2 + 1) = -a^3 + a + 1 - a^{-2},$$

which is the same. For  $a > 1$  both factors  $(a^2 - 1), (a^3 - 1)$  are positive, so  $H'_a(0) < 0$ . A convex function with  $H(0) = 0$ ,  $H'(0) < 0$  and  $H(+\infty) = +\infty$  is negative on  $(0, X_a)$  and positive on  $(X_a, \infty)$  for exactly one  $X_a > 0$ : it decreases then increases, so it has exactly two zeros counted with the one at the origin. Since  $X = \pi x^2$  is increasing in  $|x|$  and  $e^X > 0$ , the last sign change of  $g_a$  is at  $|x| = \sqrt{X_a/\pi}$ .

It is natural to study the variation of  $X_a$ , first for  $a$  near 1. Putting  $a = 1 + h$ ,  $h > 0$ , one finds for fixed  $X$ , modulo  $O(h^5)$ , an expansion  $P_1 h + P_2 h^2 + P_3 h^3 + P_4 h^4$ , where

$$P_1 = 0, \quad P_2 = 2X(2X - 3), \quad P_3 = -X(2X - 3), \quad P_4 = -5X + 15X^2 - \frac{28}{3}X^3 + \frac{4}{3}X^4.$$

**OUR NOTATION** — not in the original

[C11] The original writes the exponent of the second term as  $e^{X(2h-3h^2+3h^3-4h^4)X}$ , with a doubled  $X$ , and lists the series as  $P_1 h + P_2 h^2 + P_3 h^3 + P_4 h^5 + O(h^5)$ , where  $P_4 h^4$  is meant. Both are typesetting slips; the four polynomials as printed are correct, and we have recomputed all of them. [C12]

The expression for  $P_2$  shows that for  $h$  small enough  $H_a(X) > 0$  if  $X > \frac{3}{2}$  and  $H_a(X) < 0$  if  $X < \frac{3}{2}$ . Consequently

$$\lim_{a \rightarrow 1^+} X_a = \frac{3}{2}, \quad (12)$$

which furnishes an explicit bound

$$A \leq \sqrt{\frac{3}{2\pi}}. \quad (13)$$

But this simple bound cannot be the true value of  $A$ . For  $X = \frac{3}{2}$ ,  $P_2$  and  $P_3$  vanish, and  $P_4(\frac{3}{2}) = \frac{3}{2}$ . So for  $h$  small and non-zero,  $X_a < \frac{3}{2}$ .

**EXPANSION** — a step they compressed, written out

[C13, C14] Both checked:  $\sqrt{3/(2\pi)} = 0,6909883$ , and  $P_4(3/2) = -7,5 + 33,75 - 31,5 + 6,75 = 1,5$ . The inference is that at  $X = \frac{3}{2}$  the expansion begins at order  $h^4$  with a *positive* coefficient, so  $H_a(\frac{3}{2}) > 0$  for small  $h \neq 0$ ; since  $H_a$  is negative exactly on  $(0, X_a)$ , this forces  $X_a < \frac{3}{2}$ .

Note this makes (13) a limit that is approached but not attained, and identifies the direction of improvement.

If  $a \rightarrow +\infty$  then  $X_a \rightarrow +\infty$ ; in fact  $X_a = \log a + O(1)$ .

**FILLED** — an argument omitted in the original, supplied here

[C15] The paper says “un calcul simple montre”. Here it is. In (11) with  $a$  large,  $ae^{(1-a^2)X} \rightarrow 0$  for  $X$  bounded away from 0, so the equation  $H_a(X) = 0$  reads  $e^{(1-a^2)X} = 1 + a + o(1)$ , i.e.  $X(1 - a^{-2}) = \log(1 + a) + o(1)$ , i.e.  $X_a = \log a + O(1)$  since  $1 - a^{-2} \rightarrow 1$  and  $\log(1 + a) = \log a + O(1)$ .

We have not determined the minimal value of  $X_a$ , but it is easy to estimate it semi-heuristically. The value  $a = \sqrt{2}$  gives, on setting  $q = e^{\frac{1}{2}X_a}$ ,

$$q^3 - (1 + \sqrt{2})q^2 + \sqrt{2} = 0;$$

if  $q \neq 1$ , the quadratic equation  $q^2 - \sqrt{2}q - \sqrt{2} = 0$  gives  $q = \frac{\sqrt{2}}{2}(1 + \sqrt{1 + 2\sqrt{2}})$  and

$$X_a = 2 \log q = 1,4749 \dots \quad (a = \sqrt{2}).$$

The value  $a = 2$  gives, for  $q = e^{\frac{3}{4}X}$ ,

$$q^4 - 2 \frac{q^4 - 1}{q - 1} = 0.$$

The solution  $q > 1$  is  $q = 2,9744 \dots$ , whence

$$X_a = 1,4534 \dots \quad (a = 2).$$

**EXPANSION** — a step they compressed, written out

**Both reductions, derived; both values, recomputed.** [C16, C17]

$a = \sqrt{2}$ . Then  $a^2 = 2$ ,  $a^{-2} = \frac{1}{2}$ , and (11) is  $H(X) = \sqrt{2}e^{-X} + e^{X/2} - 1 - \sqrt{2}$ . With  $q = e^{X/2}$ ,  $e^{-X} = q^{-2}$ , so  $H = 0$  reads  $\sqrt{2}q^{-2} + q - 1 - \sqrt{2} = 0$ ; multiplying by  $q^2$ ,

$$q^3 - (1 + \sqrt{2})q^2 + \sqrt{2} = 0.$$

$q = 1$  is a root ( $1 - 1 - \sqrt{2} + \sqrt{2} = 0$ ), and dividing out  $(q - 1)$  leaves  $q^2 - \sqrt{2}q - \sqrt{2} = 0$  — the step the paper omits. Its positive root is  $q = \frac{\sqrt{2}}{2}(1 + \sqrt{1 + 2\sqrt{2}})$ , and  $X = 2 \log q$ . *Recomputed:*  $q = 2,090657851$ ,  $X_a = 1,474957555$ .

$a = 2$ . Then  $a^2 = 4$ ,  $a^{-2} = \frac{1}{4}$ , and  $H(X) = 2e^{-3X} + e^{3X/4} - 3$ . With  $q = e^{3X/4}$ ,  $e^{-3X} = q^{-4}$ , so  $H = 0$  reads  $2q^{-4} + q - 3 = 0$ , i.e.  $q^5 - 3q^4 + 2 = 0$ . Factoring out  $(q - 1)$ :  $q^4 - 2q^3 - 2q^2 - 2q - 2 = 0$ . The printed form  $q^4 - 2(q^4 - 1)/(q - 1) = 0$  is the same equation, since  $(q^4 - 1)/(q - 1) = q^3 + q^2 + q + 1$ . *Recomputed:*  $q = 2,974449245$ ,  $X_a = \frac{4}{3} \log q = 1,453411859$ .

It is likely that this is roughly the optimal value accessible by this method. Indeed if one solves  $H_a(X) = 0$ , with  $H_a$  given by (11), and supposes  $a \geq 2$ , the first term is negligible. So  $X_a$  is roughly

$$\frac{\log(1 + a)}{1 - a^{-2}}.$$

The extremum of this expression is attained for  $a(a - 1) = 2 \log(1 + a)$ , which gives

$$a = 2,08137 \dots$$

EXPANSION — a step they compressed, written out

**The printed condition, and what the number is.** [C18, C19] The original prints  $a(1-a) = 2\log(1+a)$ . For  $a > 1$  the left side is negative and the right side positive, so that equation has no root there; the intended condition is  $a(a-1) = 2\log(1+a)$ , whose root is  $a = 2,081376974$  — exactly the printed value. So the equation is mistyped and the number is right.

*But it is the extremum of the surrogate, not of  $X_a$ .* Recomputing the true minimiser of  $X_a$  over  $a > 1$  on a fine grid gives  $a \approx 1,89$  with  $X \approx 1,45200$ , against  $X_2 = 1,45341$  and  $X_{2,08137} = 1,45623$ . The value 2,08137 minimises the large- $a$  approximation  $\log(1+a)/(1-a^{-2})$ , not  $X_a$  itself. The paper's hedging ("vraisemblable", "à peu près") is fair; we draw the distinction because the two numbers are easily confused.

**2.3 The correction, and Theorem 2.** In all cases the minimal value  $A(g_a)$  so obtained is not the value (2) sought. Consider indeed  $a_0$  such that  $X_0 = X_{a_0}$  is minimal, and  $H_0 = H_{a_0}$ , positive on  $[X_0, \infty)$ .

Let  $a$  (for instance near 1) be such that  $X_a > X_0$ . On  $[X_a, \infty)$ ,  $H_a$  is  $\geq 0$  and its order of growth as  $X \rightarrow +\infty$ , namely  $e^{(1-a^{-2})X}$ , is smaller than that of  $H_{a_0}$  if  $a < a_0$ . There is therefore  $T > 0$  such that  $H_{a_0} - TH_a$  is  $\geq 0$  on  $[X_a, \infty)$ . But this function is  $> 0$  on  $[X_0, X_a)$ , hence on a neighbourhood of  $X_0$ , hence for  $X \geq X'$  with  $X' < X_0$ .

The same argument applies taking any  $a_0$  such that  $X_0 < \frac{3}{2}$ . For  $a_0 = 2$  one can determine the optimal correction (corresponding to  $a$  very close to 1), giving a function  $\geq 0$  on  $[X'', \infty)$ , with

$$X'' = 1,25\dots, \quad A \leq 0,63\dots \quad (14)$$

We have made only a very rough calculation. We state the result nonetheless, to be compared with Theorem 1.

**Theorem 2.**  $A \leq 0,64$  and  $B_1 \leq 0,41$ .

EXPANSION — a step they compressed, written out

**The correction argument, unpacked.** [C20] Three claims are compressed into that paragraph.

(a)  $H_{a_0} - TH_a$  is a legitimate competitor.  $g_{a_0} - Tg_a$  is a real even  $L^1$  function, self-dual and vanishing at 0, being a linear combination of two such (§2.2). Its last sign change is at  $X$  where  $H_{a_0} - TH_a$  last changes sign, since  $G = e^{-X}H$  and  $e^{-X} > 0$ . So any  $X'$  with  $H_{a_0} - TH_a \geq 0$  on  $[X', \infty)$  gives  $A \leq \sqrt{X'}/\pi$ .

(b) *Such a  $T$  exists.* On  $[X_a, \infty)$  both  $H_{a_0}$  and  $H_a$  are  $\geq 0$ , and  $H_a > 0$  on  $(X_a, \infty)$ . As  $X \rightarrow \infty$ ,  $H_{a_0} \sim e^{(1-a_0^{-2})X}$  and  $H_a \sim e^{(1-a^{-2})X}$ ; since  $a < a_0$  we have  $1 - a^{-2} < 1 - a_0^{-2}$ , so  $H_{a_0}/H_a \rightarrow +\infty$ . The ratio is therefore bounded below by a positive number on  $[X_a, \infty)$  — it is continuous and positive on the interior and tends to  $+\infty$  at both ends of the relevant range — and  $T := \inf_{X \geq X_a} H_{a_0}(X)/H_a(X) > 0$  is the largest admissible value.

(c) *The zero moves left.* On  $[X_0, X_a)$  we have  $H_{a_0} \geq 0$  while  $H_a < 0$  (as  $X < X_a$ ), so  $H_{a_0} - TH_a > 0$  there. By continuity it stays  $> 0$  slightly to the left of  $X_0$ , so the last sign change  $X''$  satisfies  $X'' < X_0$ . That is the gain.

EXPANSION — a step they compressed, written out

**The numbers in (14), recomputed.** [C21] Taking  $a_0 = 2$  and letting  $a \rightarrow 1^+$  — the “optimal correction” the paper describes — the construction converges to

$$X'' = 1,1512203, \quad A \leq \sqrt{X''}/\pi = 0,6053468.$$

Optimising over  $a_0$  as well,  $a_0 \approx 1,5$ – $1,7$  gives  $X'' \approx 1,1210$  and  $A \leq 0,5973$ .

The printed  $X'' = 1,25\dots$ ,  $A \leq 0,63\dots$  are therefore *weaker* than what the method yields — which is exactly what the authors' own caveat says (“nous n'avons fait qu'un calcul très approché”). Theorem 2 as stated is true; we do not restate it. See §5(2).

### §3. HIGHER DIMENSIONS

We work in Euclidean  $\mathbb{R}^d$ , with  $x \cdot y = \sum_1^d x_i y_i$ ,  $\|x\| = (x \cdot x)^{1/2}$ , and

$$\widehat{f}(y) = \int f(x) e^{-2i\pi x \cdot y} dx, \quad f(x) = \int \widehat{f}(y) e^{2i\pi x \cdot y} dy. \quad (15)$$

More generally, on a Euclidean space  $E$  of dimension  $d$  with invariant measure normalised so that the cube spanned by an orthonormal basis has measure 1, and  $x \cdot y$  the inner product, (15) inverts as stated.

#### OUR NOTATION — not in the original

[D1] That normalisation of Haar measure on an *abstract* Euclidean space is stated once here and used again in §4, where  $F_\infty$  carries no preferred coordinates. We flag it because it is the hinge between §3 and §4.

Consider Fourier pairs satisfying

- 1)  $f, \widehat{f}$  real and even, not identically zero;
- 2)  $f(0) \leq 0$  and  $\widehat{f}(0) \leq 0$ ;
- 3)  $f(x) \geq 0$  for  $\|x\| \geq a_f$ ,  $\widehat{f}(y) \geq 0$  for  $\|y\| \geq a_{\widehat{f}}$ ,

put  $A(f) = \inf\{r > 0 : f(x) \geq 0 \text{ if } \|x\| > r\}$  and  $B_d = \inf A(f)A(\widehat{f})$ .

Let  $f^\#(x)$  be the (invariant) average of  $f$  over the sphere of radius  $\|x\|$ : then  $\widehat{f^\#} = (\widehat{f})^\#$ , and  $f^\#$  and  $\widehat{f^\#}$  are not zero — otherwise  $f$  and  $\widehat{f}$  would have compact support by 3). Since  $A(f^\#) \leq A(f)$  and  $A(\widehat{f^\#}) \leq A(\widehat{f})$ , one may restrict to radial pairs. Since  $(f(x/\lambda))^\wedge = \lambda^d \widehat{f}(\lambda y)$ , the argument of §1 applies and

$$B_d = A^2, \quad A = \inf A(f), \quad (16)$$

the infimum over  $f \in L^1(\mathbb{R}^d)$  radial, not identically zero, with  $f = \widehat{f}$  and  $f(0) = 0$ , having added if necessary a multiple of the radial self-dual  $\gamma(x) = e^{-\pi\|x\|^2}$ .

#### FILLED — an argument omitted in the original, supplied here

**The radialisation, both halves.** [D3] The paper compresses this to one sentence.

*Why  $f^\#$  is a competitor.* Spherical averaging commutes with the Fourier transform on radial-invariant data:  $\widehat{f^\#} = (\widehat{f})^\#$ , because the transform commutes with rotations and averaging is an integral over the rotation group. If  $f \geq 0$  for  $\|x\| \geq a_f$  then so is its spherical average, giving  $A(f^\#) \leq A(f)$ ; and  $f^\#(0) = f(0) \leq 0$ .

*Why  $f^\# \not\equiv 0$ .* Suppose  $f^\# \equiv 0$ . By 3),  $f \geq 0$  on  $\{\|x\| \geq a_f\}$ , and a non-negative function with vanishing spherical averages vanishes a.e. there. So  $f = 0$  a.e. on  $\{\|x\| \geq a_f\}$ :  $f$  has compact support. The same argument applied to  $\widehat{f}$  gives  $\widehat{f}$  compactly supported. By Background,  $f \equiv 0$  — contradicting 1). The paper states the first half and leaves the contradiction implicit.

**EXPANSION** — a step they compressed, written out

**What changes in (16) relative to §1.2.** [D4] Only the dilation exponent:  $(f(x/\lambda))^\wedge = \lambda^d \widehat{f}(\lambda y)$ , so  $A(f) \mapsto \lambda A(f)$  and  $A(\widehat{f}) \mapsto \lambda^{-1} A(\widehat{f})$  exactly as before — the  $\lambda^d$  multiplies the *values*, not the argument, and  $A$  is unchanged by positive scalar multiples. Hence the product is again dilation-invariant, the normalisation  $A(f) = A(\widehat{f})$  is again available, and the rest of §1.2 — the admissibility of  $f + \widehat{f}$ , the compact-support contradiction, the Gaussian correction — is verbatim with  $|x|$  replaced by  $\|x\|$  and  $\gamma(x) = e^{-\pi\|x\|^2}$ .

**Theorem 3.**  $B_d \geq \frac{1}{\pi} \left( \frac{1}{2} \Gamma\left(\frac{d}{2} + 1\right) \right)^{2/d} > \frac{d}{2\pi e}$ .

*Proof.* It is modelled on the case  $d = 1$ , replacing the interval  $(-A(f), A(f))$  by the ball of centre  $O$  and radius  $A(f)$ , whose volume (which is  $\geq \frac{1}{2}$ ) is  $\frac{1}{\Gamma(\frac{d}{2}+1)} (A(f))^d \pi^{d/2}$ .  $\square$

**FILLED** — an argument omitted in the original, supplied here

**The proof, written out, and the second inequality, proved.** [D7, D8]

*First inequality.* Normalise  $\int_{\mathbb{R}^d} |f| = 1$ . As in Theorem 1:  $\int f = \widehat{f}(0) = 0$  gives  $\int f^+ = \int f^- = \frac{1}{2}$ ;  $f \geq 0$  outside the ball  $B(0, A)$  gives  $\int_{\|x\| \geq A} |f| = \int_{\|x\| \geq A} f^+ \leq \frac{1}{2}$ , hence  $\int_{B(0, A)} |f| \geq \frac{1}{2}$ ; and  $\|f\|_\infty \leq \|\widehat{f}\|_1 = 1$ . Therefore  $\text{vol } B(0, A) \geq \frac{1}{2}$ . With  $\text{vol } B(0, A) = \pi^{d/2} A^d / \Gamma(\frac{d}{2} + 1)$  this reads  $A^d \geq \Gamma(\frac{d}{2} + 1) / (2\pi^{d/2})$ , i.e.

$$B_d = A^2 \geq \frac{1}{\pi} \left( \frac{1}{2} \Gamma\left(\frac{d}{2} + 1\right) \right)^{2/d}.$$

*Second inequality.* It is equivalent to  $\left(\frac{1}{2} \Gamma(m+1)\right)^{1/m} > m/e$  with  $m = d/2$ , i.e. to  $\Gamma(m+1) > 2(m/e)^m$ . By Background,  $\Gamma(m+1) > \sqrt{2\pi m} (m/e)^m$ , so it suffices that  $\sqrt{2\pi m} \geq 2$ , i.e.  $m \geq 2/\pi = 0,6366$ , i.e.  $d \geq 4/\pi = 1,2732$  — so for every  $d \geq 2$ . For  $d = 1$  check directly:  $\Gamma(3/2) = \sqrt{\pi}/2 = 0,8862269$  and  $2(1/2e)^{1/2} = 0,8577639$ , so the inequality holds there too. The paper asserts the inequality without argument. *Verified numerically for  $d = 1, 2, 3, 4, 8, 12, 24, 48, 100$  as well.*

Setting  $X = \pi\|x\|^2$ , the argument of §2 leads to the natural functions  $g_a(x) = G_a(X)$  with  $G_a(X) = a^d e^{-Xa^2} + e^{-Xa^{-2}} - (1+a^d)e^{-X}$ ,  $H_a(X) = a^d e^{(1-a^2)X} + e^{(1-a^{-2})X} - (1+a^d)$ ,  $a > 1$ .

**EXPANSION** — a step they compressed, written out

**$g_a$  is still self-dual in dimension  $d$ .** [D9] With  $\gamma(x) = e^{-\pi\|x\|^2}$  one has  $\widehat{\gamma(a \cdot)}(y) = a^{-d} \gamma(y/a)$ , so

$$\widehat{a^d \gamma(a \cdot)} = \gamma(\cdot/a), \quad \widehat{\gamma(\cdot/a)} = a^d \gamma(a \cdot),$$

and the two outer terms of  $g_a$  swap while  $(1+a^d)\gamma$  is fixed. Hence  $\widehat{g_a} = g_a$ , and  $g_a(0) = a^d + 1 - (1+a^d) = 0$ . This is why the coefficient is  $a^d$  here where it was  $a$  in (9): it is exactly the Jacobian factor.

It is convenient to set  $a^2 = 1 + k$ ,  $d = 2c$ , whence

$$H_a(X) = (1+k)^c e^{-kX} + e^{(1-(1+k)^{-1})X} - 1 - (1+k)^c.$$

The derivative at the origin in  $X$  is  $\frac{k}{1+k} (1 - (1+k)^{c+1}) < 0$ ; the convexity argument of §2 shows that  $H_a$  has a unique positive zero  $X_a$ . As before we compute an expansion in  $k$  to order 4:  $H_a(X) = P_1 k + P_2 k^2 + P_3 k^3 + P_4 k^4 + O(k^5)$  with

$$P_1 = 0, \quad P_2 = X(X - c - 1), \quad P_3 = \frac{1}{2}(c - 2)X(X - c - 1),$$

$$P_4 = \frac{1}{12}X\{X^3 - (2c+6)X^2 + (3c(c-1)+18)X - (2c(c-1)(c-2)+12)\}.$$

EXPANSION — a step they compressed, written out

[D10, D11] The derivative at the origin: differentiating the displayed form at  $X = 0$  gives  $-k(1+k)^c + (1 - (1+k)^{-1}) = -k(1+k)^c + \frac{k}{1+k}$ , which is  $\frac{k}{1+k}(1 - (1+k)^{c+1})$ ; for  $k > 0$  and  $c > 0$  the bracket is negative. All four polynomials *recomputed and confirmed* at several  $(X, c)$ .

As in dimension 1,  $P_2$  and  $P_3$  vanish for

$$X = X(d) := \frac{d}{2} + 1. \quad (17)$$

Moreover  $P_2 > 0$  for  $X > X(d)$  and  $< 0$  for  $X < X(d)$ . Letting  $k \rightarrow 0$  we deduce  $\lim_{a \rightarrow 1} X_a = \frac{d}{2} + 1$ .

OUR NOTATION — not in the original

[D5] The original labels this equation (3.4), which it has already used for  $B_d = A^2$ . We have relabelled; the second occurrence is (17) here.

To understand the position of  $X_a$  relative to  $X(d)$  as  $a \rightarrow 1$ , compute  $Q_4(X(d))$ , where  $P_4 = \frac{X}{12}Q_4$ . The computation gives

$$Q_4(c+1) = -c^2 + 1.$$

For  $d > 2$  this term is  $< 0$ , so  $H_a(X(d)) < 0$  for  $a$  close to 1, which shows

$$X_a > \frac{d}{2} + 1 \quad (a > 1 \text{ close enough to } 1).$$

It is therefore possible that the value (17) is optimal. For  $d = 1$  it is not, as §2 showed.

EXPANSION — a step they compressed, written out

[D12, D13]  $Q_4(c+1) = -c^2 + 1$  *recomputed and confirmed* for  $c = \frac{1}{2}, 1, \frac{3}{2}, 3, 5$ . The inference: at  $X = X(d)$  the  $k$ -expansion starts at order  $k^4$  with coefficient  $\frac{X}{12}(-c^2 + 1)$ , negative for  $c > 1$  i.e.  $d > 2$ . So  $H_a(X(d)) < 0$  for small  $k > 0$ , and since  $H_a < 0$  exactly on  $(0, X_a)$ , this forces  $X_a > X(d)$ . Note this is the *opposite* direction to  $d = 1$  (where  $P_4(3/2) > 0$  forced  $X_a < 3/2$ ): in high dimension the family  $g_a$  does no better than its limit, which is why the authors suspect  $X(d)$  is optimal.

For  $d = 2$ ,  $Q_4(c+1) = 0$ , so we must compute the expansion of

$$H_a(2) = (1+k)e^{-2k} + e^{2(1-\frac{1}{1+k})} - 2 - k \quad (18)$$

to order at least 5. The Taylor expansion at 0 of  $f(z) = e^{2(1-\frac{1}{1+z})} = e^{2\frac{z}{1+z}} = \sum_0^\infty q_n z^n$  is computed by the residue theorem: putting  $w = \frac{z}{1+z}$ ,  $z = \frac{w}{1-w}$ ,  $dz = \frac{dw}{(w-1)^2}$ , one has

$$q_n = \text{Res}_{z=0} \frac{f(z)}{z^{n+1}} = \frac{1}{2i\pi} \oint e^{\frac{2z}{1+z}} \frac{dz}{z^{n+1}} = \frac{1}{2i\pi} \oint e^{2w} \frac{(1-w)^{n+1}}{w^{n+1}} \frac{dw}{(1-w)^2} = \text{Res}_{w=0} \frac{(1-w)^{n-1}}{w^{n+1}} e^{2w}.$$

EXPANSION — a step they compressed, written out

**The substitution in the residue.** [D15] With  $w = z/(1+z)$  the map is a biholomorphism near 0 fixing 0, so residues transport;  $z = w/(1-w)$  gives  $z^{-(n+1)} = (1-w)^{n+1}w^{-(n+1)}$  and  $dz = (1-w)^{-2}dw$ , whence the exponent  $n+1-2 = n-1$  in the last display. The point of the change of variable is that  $e^{2z/(1+z)}$  becomes  $e^{2w}$ , whose Taylor coefficients are immediate.

In particular  $p_5$  is the sum of

$$\frac{2^4}{4!} - \frac{2^5}{5!} \quad (19)$$

coming from the first term of (18), and of the term in  $w^5$  of  $e^{2w}(1-w)^4$ , equal to

$$\frac{2^5}{5!} - 4 \cdot \frac{2^4}{4!} + 6 \cdot \frac{2^3}{3!} - 4 \cdot \frac{2^2}{2!} + 2. \quad (20)$$

One finds  $p_5 = 0$ . Likewise  $p_6$  is the sum of

$$-\frac{2^5}{5!} + \frac{2^6}{6!} \quad (21)$$

and of

$$\frac{2^6}{6!} - 5 \cdot \frac{2^5}{5!} + 10 \cdot \frac{2^4}{4!} - 10 \cdot \frac{2^3}{3!} + 5 \cdot \frac{2^2}{2!} - 2, \quad (22)$$

whence  $p_6 = -\frac{4}{45} < 0$ . For  $a$  very close to 1 we therefore have  $H_a(2) < 0$  and  $X_a > X(2) = 2$ . Here again it is possible that the bound (17) is optimal.

**OUR NOTATION** — not in the original

[D16] The coefficients of this last expansion are called  $p_5, p_6$  while those of the earlier one were  $P_1, \dots, P_4$ . They are different objects — these are Taylor coefficients of  $H_a(2)$  in  $k$ , those were polynomials in  $X$  — and the near-identical names are the original's.

**EXPANSION** — a step they compressed, written out

[D17] *Both recomputed exactly.* (19) = 0,4 and (20) = -0,4, so  $p_5 = 0$ . (21) = -0,177 and (22) = +0,08, so  $p_6 = -0,08 = -\frac{4}{45}$ . A direct Taylor expansion of (18) confirms  $p_0 = \dots = p_5 = 0$  and  $p_6 = -4/45$ .

To conclude this section, note that for every  $d \geq 2$  we have obtained the upper bound

$$B_d \leq \mathcal{B}_d \leq \frac{d+2}{2\pi}, \quad (23)$$

where  $\mathcal{B}_d$  is defined, as in §1, by functions of  $\mathcal{S}(\mathbb{R}^d)$ . Moreover the considerations at the end of §1, on the bounds for  $L^1$  and for  $\mathcal{S}$ , apply. In the proof of (7) one must take  $T = \delta_b + \delta_{-b} + 2\delta_0$  with  $\|b\| < a = A(f)$  and  $f(b) < 0$ ;  $\hat{T} = 2\cos(2\pi b \cdot y) + 2$  is a positive plane wave. The rest of the argument is identical, replacing  $h + \hat{h}$  by the spherical average of  $h + \hat{h}$  if one wishes to stay with radial functions. In conclusion, to be compared with Theorem 3:

**Theorem 4.**

$$B_d \leq \mathcal{B}_d \leq \frac{d+2}{2\pi}, \quad B_d \geq \frac{1}{2}\mathcal{B}_d.$$

**EXPANSION** — a step they compressed, written out

[D18, D19] (23) is  $A^2 \leq X(d)/\pi = (\frac{d}{2} + 1)/\pi = (d+2)/2\pi$ , using  $\lim_{a \rightarrow 1} X_a = X(d)$  and  $A \leq A(g_a)$  for every  $a$ .

The  $d$ -dimensional version of the  $T$  argument:  $b \in \mathbb{R}^d$  is now a vector with  $\|b\| < a$ , and  $\hat{T}(y) = 2\cos(2\pi b \cdot y) + 2 \geq 0$  is constant on hyperplanes orthogonal to  $b$  — a plane wave, hence not radial, which is why the spherical average is taken at the end. Every other step of §1.4 goes through verbatim with  $|x|$  replaced by  $\|x\|$ :  $g = T * f \geq 0$  on  $\{\|x\| \geq 2a\}$  because each of  $x - b, x, x + b$  then has norm  $\geq a$ ;  $\hat{g} = \hat{T}\hat{f} \geq 0$  on  $\{\|y\| \geq a\}$ ; and the dilation  $\lambda = 1/\sqrt{2}$  balances  $2a$  against  $a$  at  $a\sqrt{2}$ . The conclusion  $B_d \geq \frac{1}{2}\mathcal{B}_d$  is (7) rearranged.



## §4. AN ARITHMETIC ARGUMENT

**CAUTION** — imported, or not fully justified here

**This section imports heavily, and we have verified none of the imports.** [E1, E5, E8, E16, E21, E23] It rests on Tate’s thesis for the functional equation (26), on Armitage for a number field with  $\zeta_F(1/2) = 0$ , on Serre for the construction of that field, on Odlyzko for discriminant lower bounds, and on Golod–Shafarevič/Brumer (via Roquette) for infinite class field towers. We state each precisely and check how the paper uses it, but we have not read those sources. Everything in §4 is therefore conditional on them in this digestion.

**4.1 Setting.** Let  $F$  be a number field of degree  $d$  over  $\mathbb{Q}$ ;  $v$  its places,  $F_v$  the completions,  $\mathcal{O}_v$  the ring of integers and  $q_v$  the residue cardinality at a finite place;  $\mathbb{A}_F = \prod'_v F_v$  the adeles,  $I_F = \mathbb{A}_F^\times$  the ideles,  $|x| = \prod_v |x|_v$  the idele norm, and

$$I_F^1 = \{x : |x| = 1\}, \quad I_F^+ = \{x : |x| \geq 1\}.$$

The measure  $dx = \prod dx_v$  is Haar; at finite places  $dx_v$  is Tate’s self-dual measure, at a real place Lebesgue, at a complex place  $dz = 2dx dy$ . At a complex place with parameters  $z = x + iy$ ,  $w = \xi + i\eta$ , Tate defines  $\widehat{f}(w) = \int f(z) e^{-2i\pi \operatorname{Tr}(zw)} dz$  with  $\operatorname{Tr}(zw) = 2 \operatorname{Re}(zw)$ .

**OUR NOTATION** — not in the original

**Reconciling the two conventions.** [E2] The paper says in one sentence that for radial functions Tate’s complex-place transform “revient à” the transform of §3 with the inner product  $z \cdot w = 2(x\xi + y\eta)$ . Spelled out:  $\operatorname{Tr}(zw) = 2(x\xi - y\eta)$  carries a minus sign, but for functions even in each variable the sign is immaterial, so one may use  $z \cdot w = 2(x\xi + y\eta)$ , which is exactly the inner product making  $dz = 2dx dy$  self-dual in the sense of §3’s normalisation. This is the point at which §3’s abstract-Euclidean convention is needed.

Let  $f$  be the Schwartz function on  $\mathbb{A}_F$

$$f(x) = \prod_{v|\infty} f_v(x_v) \prod_{v \text{ finite}} f_v^0(x_v) \quad (24)$$

with  $f_v^0 = \mathbf{1}_{\mathcal{O}_v}$ . The associated Tate zeta function is  $Z(f, s) = \int_{I_F} f(x) |x|^s d^\times x$  for  $\operatorname{Re}(s) > 1$ . Rather than the decomposable functions of (24) we consider functions

$$f(x) = f_\infty(x_\infty) \prod_{v \text{ finite}} f_v^0(x_v), \quad f_\infty \in \mathcal{S}(\mathbb{R}^d), \quad (25)$$

where  $\mathbb{R}^d = F_\infty$  is Euclidean via  $\|x_\infty\|^2 = \sum_{v \text{ real}} |x_v|^2 + \sum_{v \text{ complex}} 2\|z_v\|^2$ . Tate’s conditions  $(z_1), (z_2), (z_3)$  hold for such functions. By Tate,

$$Z(f, s) = \int_{I_F^+} f(x) |x|^s d^\times x + \int_{I_F^+} \widehat{f}(x) |x|^{1-s} d^\times x + \kappa \frac{\widehat{f}(0)}{s-1} - \kappa \frac{f(0)}{s}, \quad (26)$$

where  $\kappa = \frac{2^{r_1} (2\pi)^{r_2} h R}{\sqrt{D_F} w}$  is the residue of  $\zeta_F$  at  $s = 1$ ,  $D_F$  the absolute discriminant,  $d = r_1 + 2r_2$ . Both integrals in (26) converge absolutely for all  $s \in \mathbb{C}$ .

#### 4.2 The two lemmas.

**Lemma 1.** *Let  $s$  be a zero of  $\zeta_F(s)$  with  $\operatorname{Re}(s) > 0$ . Then  $Z(f, s)$  vanishes at  $s$ , for every choice of  $f_\infty \in \mathcal{S}(F_\infty)$ .*

Indeed one may write, first for  $\operatorname{Re} s > 1$ ,  $Z(f, s) = Z(f_\infty, s)\zeta_F(s)$ . Since  $Z(f, s)$ ,  $\zeta_F(s)$  and  $Z(f_\infty, s)$  are holomorphic for  $s \neq 1$ ,  $\operatorname{Re}(s) > 0$ , the lemma follows.

**FILLED** — an argument omitted in the original, supplied here

**The factorisation and the continuation.** [E7]

*The factorisation.* For  $\operatorname{Re} s > 1$  the idele integral factors over places because  $f$  does:  $Z(f, s) = \prod_v Z(f_v, s)$ . The finite places contribute  $\prod_{v \text{ finite}} (1 - q_v^{-s})^{-1} = \zeta_F(s)$ , since  $f_v^0 = \mathbf{1}_{\mathcal{O}_v}$  and the local zeta integral of the characteristic function of the integers is the local Euler factor (with  $d^\times x_v$  normalised by the factor  $(1 - q_v^{-1})^{-1}$  that the paper records). The archimedean places contribute  $Z(f_\infty, s)$  by definition. Hence  $Z(f, s) = Z(f_\infty, s)\zeta_F(s)$  on  $\operatorname{Re} s > 1$ .

*The continuation.*  $Z(f, s)$  extends holomorphically to  $\{\operatorname{Re} s > 0\} \setminus \{1\}$  by (26), whose two integrals are entire.  $Z(f_\infty, s)$  is holomorphic there because  $f_\infty \in \mathcal{S}$  makes  $\int_{F_\infty} f_\infty \prod |x_v|^{\sigma-1} dx$  absolutely convergent for  $\sigma > 0$  — which is the paper's remark about condition (z<sub>3</sub>). Two holomorphic functions agreeing on  $\operatorname{Re} s > 1$  agree on the connected set  $\{\operatorname{Re} s > 0\} \setminus \{1\}$ , so the identity persists, and at a zero  $s$  of  $\zeta_F$  with  $\operatorname{Re} s > 0$  we get  $Z(f, s) = 0$ .

For every finite place,  $\widehat{f}_v^0 = |\mathfrak{d}_v|^{-1/2} \mathbf{1}_{\mathfrak{d}_v^{-1}}$ , where  $\mathfrak{d}_v$  is the different; and  $\prod_{v \text{ finite}} |\mathfrak{d}_v| = |D_F|$ . Consider the first integral of (26). If  $f(x) \neq 0$  at  $x = (x_\infty, x_f)$  then the description of  $f_f$  gives  $|x_f| \leq 1$ ; since  $|x_\infty x_f| \geq 1$  on  $I_F^+$ ,

$$|x_\infty| = \prod_{v|\infty} |x_v| \geq 1. \quad (27)$$

In the second integral, noting  $|x_v| \leq |\mathfrak{d}_v|$  if  $x_v \in \mathfrak{d}_v^{-1}$ , one has  $|x_f| \leq \prod_v |\mathfrak{d}_v| = |D_F|$ , whence

$$|x_\infty| \geq D_F^{-1}. \quad (28)$$

**EXPANSION** — a step they compressed, written out

**The two inferences** (27)–(28). [E9] On  $I_F^+$  the idele norm satisfies  $|x| = |x_\infty| |x_f| \geq 1$ . In the first integral  $f \neq 0$  forces  $x_v \in \mathcal{O}_v$  at every finite place, so  $|x_v|_v \leq 1$  and  $|x_f| \leq 1$ ; combined with  $|x_\infty| |x_f| \geq 1$  this gives  $|x_\infty| \geq 1$ . In the second,  $\widehat{f} \neq 0$  forces  $x_v \in \mathfrak{d}_v^{-1}$ , where  $|x_v|_v \leq |\mathfrak{d}_v|$ , so  $|x_f| \leq \prod_v |\mathfrak{d}_v| = |D_F|$  and therefore  $|x_\infty| \geq |D_F|^{-1}$ . The asymmetry between the two — 1 against  $D_F^{-1}$  — is the entire arithmetic input of §4: it is what puts the discriminant into Proposition 1.

**Lemma 2.** *Suppose there is a Fourier pair  $(f, \widehat{f})$  on  $F_\infty = \mathbb{R}^d$  such that  $f(x_\infty) \geq 0$  if  $|x_\infty| \geq 1$ ,  $f$  takes strictly positive values in a neighbourhood of 1 inside  $\{|x_\infty| \geq 1\}$ ,  $\widehat{f}(y_\infty) \geq 0$  if  $|y_\infty| \geq D_F^{-1}$ , and  $f(0) = \widehat{f}(0) = 0$ . Then  $\zeta_F(s) \neq 0$  for every  $s$  in the interval  $]0, 1[$ .*

Indeed (26) is then reduced to its integral terms;  $|x|^s$  is strictly positive on the domain of integration, and the integral (27) is strictly positive by the property imposed on  $f$ . So  $Z(f, s) > 0$  and  $\zeta_F(s) \neq 0$  by Lemma 1.

**EXPANSION** — a step they compressed, written out

**Which hypothesis does what.** [E11]

- $f(0) = \widehat{f}(0) = 0$  kills the two polar terms  $\kappa \widehat{f}(0)/(s-1)$  and  $-\kappa f(0)/s$  of (26). Without this,  $Z(f, s)$  would have a pole at  $s = 1$  and no sign could be read off.
- $f \geq 0$  on  $\{|x_\infty| \geq 1\}$  makes the first integral  $\geq 0$ : by (27) the integrand is supported where  $|x_\infty| \geq 1$ .

- $\hat{f} \geq 0$  on  $\{|y_\infty| \geq D_F^{-1}\}$  makes the second integral  $\geq 0$ , by (28).
- “ $f$  takes strictly positive values near  $|x_\infty| = 1$ ” upgrades the first integral from  $\geq 0$  to  $> 0$ . This is the only hypothesis that gives strictness, and strictness is the whole point:  $Z(f, s) > 0$  contradicts  $Z(f, s) = 0$ .
- $s \in ]0, 1[$  real makes  $|x|^s > 0$ ; for complex  $s$  no sign argument is available, which is why the conclusion concerns only real zeros.

Let  $x = (x_v) \in F_\infty$ . The Euclidean norm compatible with Tate’s Fourier transform is  $\|x\|^2 = \sum_{v \text{ real}} |x_v|^2 + 2 \sum_{v \text{ complex}} \|x_v\|^2$ . Since  $|x|^2 = \prod_{v \text{ real}} |x_v|^2 \prod_{v \text{ complex}} \|x_v\|^4$ , the arithmetic–geometric inequality gives

$$|x|^{2/d} \leq \frac{1}{d} \|x\|^2.$$

Putting  $r = \|x\|$ ,  $\rho = \|y\|$ , one sees that

$$|x| \geq 1 \implies r \geq \sqrt{d}, \quad |y| \geq |D_F|^{-1} \implies \rho \geq |D_F|^{-1/d} \sqrt{d}.$$

**EXPANSION** — a step they compressed, written out

**The AM–GM step, with the places counted.** [E12, E13] There are  $r_1$  real and  $r_2$  complex places,  $d = r_1 + 2r_2$ . Write the  $d$  non-negative quantities

$$\underbrace{|x_v|^2}_{v \text{ real, } r_1 \text{ terms}}, \quad \underbrace{\|x_v\|^2, \|x_v\|^2}_{v \text{ complex, } 2r_2 \text{ terms}}$$

— each complex place contributing its  $\|x_v\|^2$  *twice*. Their sum is exactly  $\|x\|^2$  and their product is exactly  $|x|^2$ . AM–GM on  $d$  terms gives  $(|x|^2)^{1/d} \leq \frac{1}{d} \|x\|^2$ , which is the display. The factor 2 in  $\|x\|^2$  and the fourth power in  $|x|^2$  are the same phenomenon counted once each, which is easy to mis-count.

Then  $|x| \geq 1$  gives  $\|x\|^2 \geq d|x|^{2/d} \geq d$ , i.e.  $r \geq \sqrt{d}$ ; and  $|y| \geq |D_F|^{-1}$  gives  $\rho^2 \geq d|D_F|^{-2/d}$ , i.e.  $\rho \geq |D_F|^{-1/d} \sqrt{d}$ .

**Proposition 1.** *Suppose there is a number field  $F$  of degree  $d$  and discriminant  $D$  such that  $\zeta_F(s)$  has a zero in  $]0, 1[$ . Then  $\mathcal{B}_d \geq d|D|^{-1/d}$ .*

Conversely, of course,  $\zeta_F$  has no real zero if  $d|D|^{-1/d} > \mathcal{B}_d$ .

**FILLED** — an argument omitted in the original, supplied here

**The proof, which the paper calls evident.** [E14, E15] Suppose  $d|D|^{-1/d} > \mathcal{B}_d$ . By definition of  $\mathcal{B}_d$  as an infimum over Schwartz competitors, and by the radial reduction of §3, there is a radial Schwartz Fourier pair  $(f, \hat{f})$  with  $f = \hat{f}$  up to the dilation of §1.2,  $f(0) = \hat{f}(0) = 0$ , and

$$A(f)A(\hat{f}) < d|D|^{-1/d}.$$

Dilating (which preserves the product) we may arrange  $A(f) \leq \sqrt{d}$  and  $A(\hat{f}) \leq |D|^{-1/d} \sqrt{d}$ , i.e.  $f \geq 0$  for  $r \geq \sqrt{d}$  and  $\hat{f} \geq 0$  for  $\rho \geq |D|^{-1/d} \sqrt{d}$ . By the two implications just displayed, these are exactly the hypotheses of Lemma 2 in terms of the idele norm. One may further assume  $f > 0$  on  $\{\sqrt{d} \leq \|x\| \leq \sqrt{d} + \varepsilon\}$  — otherwise  $f$  vanishes on an open set beyond its last sign change, and one shrinks  $A(f)$  — which supplies the strictness hypothesis, since  $\|\mathbf{1}\| = \sqrt{d}$  for the idele  $\mathbf{1}$ . Lemma 2 then gives  $\zeta_F \neq 0$  on  $]0, 1[$ . The proposition is the contrapositive.

**4.3 Fields with a real zero, and linear growth.** It is difficult to find fields  $F$  satisfying the hypothesis of the proposition. However, the decomposition, for  $F$  Galois over  $E$ , of  $\zeta_F$  into Artin  $L$ -functions for  $E$  allowed Armitage to exhibit such a zero (at  $s = \frac{1}{2}$  of course, in accordance

with the Riemann hypothesis). More precisely, Armitage considers an explicit extension  $F$  of  $E = \mathbb{Q}(\sqrt{3(1+i)})$ , of degree 12 over  $E$  and hence of degree 48 over  $\mathbb{Q}$ , constructed by Serre, and shows that  $\zeta_F(\frac{1}{2}) = 0$ . Consequently number theory implied a priori the following weak version of Theorem 3:

**Proposition 2.** *If  $d$  is a multiple of 48, then  $\mathcal{B}_d$  is strictly positive.*

For  $d = 48$  this follows from the existence of  $F$ . Suppose  $d = 48c$ . There is a cyclotomic extension  $L$  of  $\mathbb{Q}$  of degree  $c$  linearly disjoint from  $F$ . Then  $LF$  has degree  $d$  over  $\mathbb{Q}$ , and  $\zeta_F$  divides  $\zeta_{LF}$  since  $LF/F$  is abelian and  $\zeta_{LF}$  therefore factors into Dirichlet  $L$ -functions relative to  $F$ .

EXPANSION — a step they compressed, written out

**Two silent uses.** [E18] (a) A cyclotomic field of each degree, disjoint from  $F$ . The cyclotomic fields  $\mathbb{Q}(\zeta_n)$  have degrees  $\varphi(n)$ , which do not take every value; but one may take a subfield of  $\mathbb{Q}(\zeta_p)$  for a large prime  $p \equiv 1 \pmod{c}$ , whose Galois group is cyclic of order  $p-1$ , and hence has a subfield of degree exactly  $c$ . Choosing  $p$  unramified in  $F$  makes  $L$  linearly disjoint from  $F$ . (b)  $\zeta_F \mid \zeta_{LF}$ . For  $LF/F$  abelian with group  $G$ ,  $\zeta_{LF} = \prod_{\chi \in \hat{G}} L(s, \chi)$  over  $F$ , and the trivial character contributes  $\zeta_F$ . So the quotient is a product of  $L$ -functions, entire, and every zero of  $\zeta_F$  is a zero of  $\zeta_{LF}$ . Proposition 1 applied to  $LF$  then gives  $\mathcal{B}_d > 0$ .

One may ask whether Proposition 1 implies a restriction on the discriminants of fields whose  $\zeta_F$  has a real zero. In that case

$$|D|^{1/d} \geq \frac{d}{\mathcal{B}_d}. \quad (29)$$

But unconditionally, by Theorem 3,  $\frac{d}{\mathcal{B}_d} < 2\pi e = 17,079 \dots$ , whereas Odlyzko's lower bounds give in general  $|D|^{1/d} \geq 22,2(1+o(1))$  as  $d \rightarrow \infty$ . Consequently (29) is automatically satisfied, at least for  $d$  large enough.

EXPANSION — a step they compressed, written out

[E20, E21, E22] The chain is  $\mathcal{B}_d \geq B_d \geq d/(2\pi e)$  from (5) and Theorem 3, so  $d/\mathcal{B}_d \leq 2\pi e = 17,0795$ . Since Odlyzko's constant 22,2 exceeds it, (29) carries no information. The original prints Odlyzko's bound as  $22,2(1+o(d))$ ;  $o(1)$  is meant. This is a *negative* result, and the authors include it deliberately: Proposition 2 does not improve on Theorem 3. What follows is the part that does say something.

Proposition 2 therefore does not lead to an interesting lower bound for  $\mathcal{B}_d$ . It is striking to note nonetheless that, for certain degrees at least, number theory implied the linear growth in  $d$  given by Theorem 3.

Let  $p$  be a prime. By the theorems of Golod–Shafarevič and Brumer there is a sequence of fields  $E_p^1 \subset E_p^2 \subset \dots$  where  $E_p^1$ , of degree  $p(p-1)$  over  $\mathbb{Q}$ , is an extension of degree  $p$  of  $\mathbb{Q}(\zeta_p)$ , and  $E_p^{n+1}/E_p^n$  is abelian, unramified, of degree  $p$ . Consider the extensions  $F_i = F E_p^i$  of  $F$ ; extracting a strictly increasing subsequence gives  $F_0 \subset F_1 \subset \dots \subset F_m \subset \dots$ , each extension abelian of degree  $p$ . In the absence of relative ramification, a classical formula gives

$$D(F_m) = D(F_0)^{p^m} := D^{p^m}. \quad (30)$$

The successive extensions from  $F$  being abelian,  $\zeta_F$  divides  $\zeta_{F_m}$  for all  $m$ . Proposition 1 then gives, for  $d = d_0 p^m$  with  $d_0 = [F_0 : \mathbb{Q}]$ ,

$$\mathcal{B}_d \geq C d, \quad C = |D|^{-1/d_0}. \quad (31)$$

For such sequences of degrees, (23) and (31) therefore show that the growth of  $\mathcal{B}_d$  — and hence of  $B_d \geq \frac{1}{2}\mathcal{B}_d$  — is linear in  $d$ .

EXPANSION — a step they compressed, written out

**Why (31) follows, and two printed slips.** [E24, E25, E26] By Proposition 1 applied to  $F_m$ ,  $\mathcal{B}_d \geq d|D(F_m)|^{-1/d}$  with  $d = d_0 p^m$ . By (30),  $|D(F_m)| = |D|^{p^m}$ , so  $|D(F_m)|^{-1/d} = |D|^{-p^m/(d_0 p^m)} = |D|^{-1/d_0}$ , a constant independent of  $m$ . Hence  $\mathcal{B}_d \geq Cd$  with  $C = |D|^{-1/d_0}$  — the point being that the discriminant grows exactly fast enough for the bound to stay linear rather than decay.

The original typesets the exponent as  $|D|^{-1/d_0}$ ; and the sentence after (31) cites “(3.10) et (4.8)”, where the pair actually being combined is the upper bound (23) and the lower bound (31).

If  $p$  does not divide  $D_F$ , then  $F$  and  $\mathbb{Q}(\zeta_p)$  are linearly disjoint and one may choose  $E_p^1$  linearly disjoint from  $F$ . Then  $F_0 = F E_p^1$  and the inequality is valid for  $d = 48(p-1)p^n$ ,  $n \geq 1$ . Of course the factor  $(p-1)$  is not necessary if one is willing to use Artin’s conjecture or Dedekind’s divisibility conjecture.

EXPANSION — a step they compressed, written out

**The point of §4, which is easy to lose.** [E27] Theorem 3 gives  $B_d > d/(2\pi e)$  by a two-line argument about the volume of a ball. §4 shows that, along an infinite sequence of degrees, the *same* linear growth is a consequence of number theory: a field whose zeta function has a real zero forces  $\mathcal{B}_d \geq d|D|^{-1/d}$ , and class field towers produce fields of unboundedly large degree whose root discriminants stay bounded. So the linear growth in Theorem 3 is not an artefact of the crude volume bound — it is the truth, and it was implicit in known facts about ramification before anyone asked the Fourier question.

## §5. EDITORIAL REMARKS — EVERYTHING LATER THAN THE PAPER

Quarantined here; nothing above depends on any of it.

- (1) **The Hermite bound is 3/2, not 3.** [c2] §2.1 carries out the computation the paper describes but does not perform. With the paper’s own Fourier convention the unique (up to scale) self-dual combination of  $H_0, H_2, H_4$  with  $f(0) = 0$  is  $f(x) = x^2(x^2 - \frac{3}{2\pi})e^{-\pi x^2}$ , whose last sign change is at  $\sqrt{3/(2\pi)}$ , giving  $\pi A^2 = 3/2$ . The printed  $\pi A^2 \leq 3$  is true but weaker by a factor 2, and 3/2 is exactly the bound (2.5) obtained later from the  $g_a$  family. So the two routes of §2 — Hermite, and the Gaussian family — give the *same* bound, which the paper does not remark on. We have not resolved whether the printed 3 is a slip or reflects a different normalisation of  $H_4$ .
- (2) **Theorem 2 is weaker than its own method.** [c21] The correction argument of §2.3, optimised numerically, gives  $X'' = 1,1512$  and  $A \leq 0,6053$  with  $a_0 = 2$ , and  $X'' \approx 1,1210$ ,  $A \leq 0,5973$  optimising  $a_0$  as well; the paper states  $X'' = 1,25\dots$ ,  $A \leq 0,63\dots$  and Theorem 2 rounds to  $A \leq 0,64$ . The authors flag their computation as “très approché”, so this is a sharpening of an acknowledged estimate, not a correction. We state Theorem 2 as they do.
- (3) **Where the true minimiser sits.** [c19]  $a = 2,08137$  minimises the large- $a$  surrogate  $\log(1+a)/(1-a^{-2})$ ; the true minimiser of  $X_a$  is  $a \approx 1,89$ , with  $X \approx 1,45200$  against 1,45341 at  $a = 2$  and 1,45623 at  $a = 2,08137$ . All three are worse than the corrected value of item 2, which is the point of §2.3.
- (4) **What the paper leaves open.** Two things, both explicit:

- The exact value of  $B_1$ . The paper brackets it as  $0,1687 \leq B_1 \leq 0,41$  and determines nothing further. The gap is a factor of about 2,4.
- Whether  $X(d) = \frac{d}{2} + 1$  is optimal for  $d \geq 2$ , i.e. whether  $\mathcal{B}_d = (d+2)/2\pi$ . The paper says twice that this “is possible”, having shown that the family  $g_a$  cannot beat it for  $d > 2$  (via  $Q_4(c+1) < 0$ ) and for  $d = 2$  (via  $p_6 < 0$ ) — in contrast with  $d = 1$ , where it can.

Both are recorded on the project’s Open problems page with what we could verify of their status.

- (5) **The linear growth constant is now known, and both of their bounds were within a factor of two.** Theorem 3 and Theorem 4 bracket  $B_d$  between  $d/2\pi e$  and  $(d+2)/2\pi$ , a factor  $e$  apart. In August 2026 — sixteen years later — Chapter 1 of OpenAI’s *Ten Advances in Mathematics and Theoretical Computer Science* states, as its Theorem 1.2, that

$$\lim_{d \rightarrow \infty} \frac{A_+(d)}{\sqrt{d}} = \frac{1}{\pi}, \quad \text{whence} \quad \frac{B_d}{d} \rightarrow \frac{1}{\pi^2} = 0,101321\dots$$

where  $A_+(d)$  is exactly the  $A$  of (16). The value sits inside their bracket, a factor  $\pi/2 = 1,5708$  below their upper constant  $1/2\pi = 0,159155$  and a factor  $2e/\pi = 1,7305$  above their lower constant  $1/2\pi e = 0,058550$ . So the crude volume bound of Theorem 3 and the Gaussian family of Theorem 4 were each wrong by less than a factor of two, and the truth lies between them, nearer the upper end.

Two things must be said about that reference. Its results are stated to have been produced by an internal OpenAI model, and the manuscript is self-published rather than refereed. And we have read its statements, not its proof — so this remark is reported, not verified, exactly like the imports of §4. The same source’s Theorem 1.2 also settles the question of item 4 in the negative for large  $d$ : since  $1/\pi^2 < 1/2\pi$ , the value  $X(d) = \frac{d}{2} + 1$  is *not* asymptotically optimal. Cohn–Gonçalves had already shown it is not optimal at  $d = 12$ , where the sharp value is  $B_{12} = 2$  against their  $(12+2)/2\pi = 2,2282$ .

For  $d = 1$  none of this helps: the exact value of  $B_1$  is open, and Cohn–Gonçalves record that no sharp value is known or even conjectured in any dimension but 12.

- (6) **The source.** The authors’ arXiv LaTeX does not build on a 2026 distribution: [français] must become [french], and two stray 0xCA bytes (Mac Roman non-breaking spaces, at lines 58 and 400) must be replaced. With those two changes it compiles clean to exactly 19 pages, matching the arXiv PDF. The file is ISO-8859, not UTF-8. [F2, F3]